ARTS Course Modernization Using Apriori and Association Rules

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***Abstract*—Providing relevant and engaging courses is now more important than ever in higher education as universities in general are facing a plateau in enrollment. There is a high need in higher education for a strategic overhaul of curriculum materials and for working to align courses with current modernization standards and student interest and investment. Modern data mining approaches are a viable way to identify relevant courses and electives based on student participation.**

Keywords—modernization, enrollment, electives, data mining

I. INTRODUCTION

Like other university programs across the globe, the College of Arts and Sciences at the University of Diploma Printing in Romanigstan is facing a plateau in enrollment as student engagement and interests shift. Specifically, the Department of Arts recognized the need for changes to its Bachelor of Fine Arts (BFA) curriculum to aim for a broader appeal to students. By using data mining approaches, this paper aims to identify the more relevant electives and to help narrow the focus of the BFA program.

A. Problem Statement

The BFA curriculum has not significantly evolved over the past five years, leading to stagnant enrollment. The current curriculum comprises 39 credit hours, which include seven core and six elective courses. These courses may no longer align with the changing interests of the student body or the latest trends in this area of study.

B. Program Objectives

1) Refine credits: The BFA program has a goal to reduce the number of credit hours from 39 to 33 to help make the program more accessible to students.

2) Align with student interest: The BFA program wants to evaluate its elective offerings based on student interest over the past five years.

3) Justify sustainability: The BFA program looks to determine a rationale for which courses need to remain based on trends in course enrollment.

C. Methodology

Following the CRoss Industry Standard Process for Data Mining (CRISP-DM) approach, as outlined in [1], and using algorithms that identify patterns of frequency that make use of a threshold to trim off less frequent elective choices, will help us accomplish this task. The CRISP approach has a series of steps that include data cleaning and data preparation that are further used to enhance the understanding of the business objectives and incorporate new information into the data mining process and modeling process. This approach is designed to work around biases and make use of discoveries.

As for the data mining processes, we make use of KNIME [2] for initial data cleaning steps and analysis, and for the more complex algorithms such as Apriori for examining frequency patterns in elective choices and association rules for further refinement.

II. DESCRIPTION OF THE DATASET AND INITIAL PREPROCESSING

A. Datset Description

An initial look at the data reveals a five-year sample of course enrollment at the University of Diploma Printing in Romanigstan. The data provides us with the student's name, the term the course was taken and a description of the course.

We do have a separate list of Arts courses that define the program that are in a document format as well. They include the type of course and help us identify the electives we are studying, and they include the course numbers and descriptions that appear to match the descriptions in our data file.

To understand the data, let's look at our files in detail. First, there is the Projectdata.csv file:

- It has the following columns: "Student name" which is the name of the student taking the course; "semester new" which is the term and year of the course; and "coursename," which is the title and description of the course.
- There are 4,917 rows of data.
- For unique values, "Student name" has 448, "semester new" has 16, and "coursename" has 169.

Looking at the list of arts courses after converting it into a csv file, this is what we see:

- It has the following columns: "Course Number," which identifies the catalog listing of the course; "Course Name," which identifies the course specifically by a descriptive name; and "Course Type," which will help us identify our electives as part of our study.
- There are 42 rows.
- For unique values, "Course Number" has 42, "Course Name" has 41, and "Course Type" has 3.

A deeper look at the Projectdata.csv file reveals the following issues with it:

- *Missing course name values:* There are three rows missing "coursename" values.
- *Duplicate rows:* There are 1,245 rows with duplicate information where the "Student name," "semester new," and "coursename" are all the same for at least

one other record. An example of this is seen in repeated records for "Bill Mumy, Spring 2003, CONTEMP ART – since 1945."

- *Inconsistencies in course names:* With the "coursename" field. Using Jaccard similarity scores of 70% or higher reveal course names that are quite similar, such as "Contemp Art – 1945 to Today," and "Contemp Art – 1945 to Present." Overall, there are four similar course names that score above 70% in terms of Jaccard similarity matching in their descriptions.
- *Non*-Arts courses: In the file, there are 134 unique "coursenames" that don't match back to the Arts curriculum courses.

In the data, student names, however, appear to be unique. Performing a Jaccard similarity does not find any significant similarities with the names of the students.

A deeper look at the course list file reveals that there is one duplicate "Course Name" that has two listings and "Course Number" values, which is "FRANCE & THE EUROP.UNION." There do not appear to be any other significantly similar course names, as ones with a high Jaccard similarity have a clear semantic difference, such as "American health policy," compared to "American social policy."

B. Initial Preprocessing

Initial preprocessing is straightforward, keeping in mind that we must solve some larger issues. First steps include:

- Create a comma separated variable file for the Arts course offerings based on the document.
- Import both the main student data file and the course offerings into our KNIME project.
- Remove the null records.
- Remove the duplicates in the student data file.
- Remove the records with blank descriptions.
- Examine similarities in course description using a distance measure such as Jaccard distance with an applied threshold such as 70% and determine if we want to merge those records.
- Match the student data file course descriptions to the course name descriptions in the course offerings file using a join node in KNIME. The process will weed out the courses that are not from the Arts program and we can isolate them to just the electives that we are examining.
- Create charting and descriptive statistics to get an idea of distribution frequencies, measures of central tendency and range of the data.

C. Additional Issues to Resolve

These initial steps still do not address what to do with the one significant issue with two courses with different course numbers in the course list having the exact same course name. If we are matching based on course name descriptions, we may have to consolidate the two courses into one set of records unless that distinction can be made, and from our data sample, that distinction does not appear to be possible.

III. PREPROCESSING

The preprocessing steps revealed that preparing our data was not as simple as running Jaccard similarity and fixing those with higher scores as some of the differences in course names were very subtle, with the difference being a single character at times. Furthermore, the preprocessing involved some judgment calls as to how to best merge the course names, and illustrated the importance of CRISP-DM as further conversations with the clients are warranted to ensure that the data is tuned for analysis.

A. Preprocessing Steps

Using both Excel and KNIME, the Projectdata.csv file was processed with the following steps:

- Removal of null records in KNIME using the "Row Filter" node. This took the count of rows from 4,917 to 4,914.

Table "default" - Rows: 4914 | Spec - Columns: 3 | Properties | Flow Variables

Row ID	Studen...	semest...	coursename
Row0	Bill Mumy	Fall 2004	BEHAVIORAL PHARMACOLOGY
Row1	Bill Mumy	Fall 2000	AMERICAN FOREIGN POLICY
Row2	Bill Mumy	Fall 2003	DRUGS, BRAIN AND MIND
Row3	Bill Mumy	Fall 2005	Environmental Case Studies

Fig 1. KNIME row count for removing blank records.

- Consolidation of duplicates in KNIME using the "Duplicate Row Filter." The resulting set weeded out many records, leaving us with 3,669 rows.

Table "default" - Rows: 3669 | Spec - Columns: 3 | Properties | Flow Variables

Row ID	Studen...	semest...	coursename
Row0	Bill Mumy	Fall 2004	BEHAVIORAL PHARMACOLOGY
Row1	Bill Mumy	Fall 2000	AMERICAN FOREIGN POLICY
Row2	Bill Mumy	Fall 2003	DRUGS, BRAIN AND MIND
Row3	Bill Mumy	Fall 2005	Environmental Case Studies
Row4	Bill Mumy	Fall 2000	COMPUTER LINEAR ALGEBRA

Fig 2. KNIME row count for removing duplicates.

- Standardization of course names and student names. I decided to use the course list file as the key for standardizing the course names in the project data file. This process included manual inspection of the Projectdata.csv course values by sorting them in Excel and charting the differences. Ultimately, a large series of replace functions were put in place in KNIME using the "String Manipulation" node. I identified around 40 replacements, followed by a couple of additional replacement functions to ensure the course names were standardized to get the desired result and had to further tune the code for accidental appended subsets due to the replacement being a pure string within the string. Many student names also went through a similar replacement process due to what appeared to be a data problem with an additional leading capital letter, as well as some standardization of titles like "Senator" and "Governor." Many student names also had a leading letter, which appeared to be some sort of data

extraction error with the original file, and some had numerals and I also did some standardization with suffixes like "Jr." After capitalization standardization and the join between files for the course name matches, an additional cleanup was performed to make a few of the course names more intelligible.

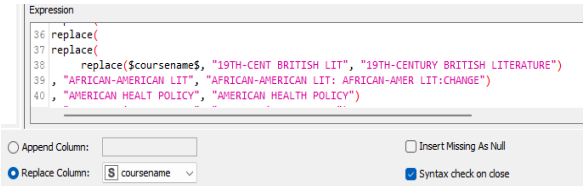


Fig 3. Replace function code in KNIME.

- Removal of courses not offered by the program. Using a KNIME "Joiner" node, the "coursename" field was joined to course name information in the course list file. One step I did prior was to consolidate the two "FRANCE & THE EUROP.UNION" listings into one record, and I set the course as "ARTS 571 & 577" with the purpose of following up with the client and discuss if that course is just one course or two. The join resulted in 2,161 records, with possible adjustments to that count based on conversations with the client over course name replacements and logical decisions on how those courses were renamed and consolidated.

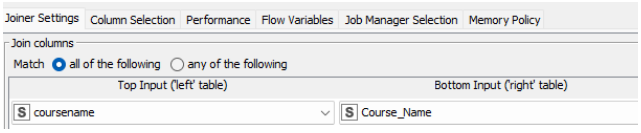


Fig 4. Join column variable selection for project data and course list.

- Row filtering. I used a "Row Filter" node to include rows only where the type of course was an elective since those are the courses we are examining as part of our data mining exercise. That took the count down to 700 records.
- Column filtering. Using a column filter, we can further refine our result to contain the student name, the semester information, the course number, and the course name.

Table "default" - Rows: 700 | Spec - Columns: 4 | Properties | Flow Variables

Row ID	Studen...	semest...	Course...	Course_Name
Row27_Row23	Chris Evert	Spring 2001	ARTS 549	Freshwater Ecology
Row28_Row7	Chris Evert	Spring 2002	ARTS 491	Contemporary Pol..
Row29_Row9	Chris Evert	Spring 2003	ARTS 493	American Health P..
Row30_Row23	Ruth Bader ...	Spring 2002	ARTS 549	Freshwater Ecology
Row31_Row7	Ruth Bader ...	Spring 2001	ARTS 491	Contemporary Pol..
Row32_Row9	Ruth Bader ...	Spring 2001	ARTS 493	American Health P..

Fig 5. Refined columns for our analysis.

- Duplicate row filtering again. One last duplicate row filter was put in place to ensure that our joined and cleaned records did not contain any duplicates, and one additional record was removed due to this action, leaving us with 699 records.

B. Preprocessing Observations

Immediately, a few courses emerge as candidates for removal as they have zero matches. Those courses include: "AMERICAN SOCIAL POLICY," "EARLY BALCAN HIST/SOC," and "French Thought Till 1945." Results from the data cleanup are already fruitful in our task. We have 699 remaining electives. We now have our full preprocessing result, and the basis for our full model.

Table "default" - Rows: 699 | Spec - Columns: 4 | Properties | Flow Variables

Row ID	Studen...	semest...	Course...	Course_Name
Row27_Row23	Chris Evert	Spring 2001	ARTS 549	Freshwater Ecology
Row28_Row7	Chris Evert	Spring 2002	ARTS 491	Contemporary Pol...
Row29_Row9	Chris Evert	Spring 2003	ARTS 493	American Health P...
Row30_Row23	Ruth Bader ...	Spring 2002	ARTS 549	Freshwater Ecology
Row31_Row7	Ruth Bader ...	Spring 2001	ARTS 491	Contemporary Pol...
Row32_Row9	Ruth Bader ...	Spring 2001	ARTS 493	American Health P...

Fig 6. After filtering for only electives, we now have 699 rows for further analysis and modeling.

At this state, we can take that result and consider it ready for modeling steps using Apriori with association rules to help us find the frequent itemsets, considering each student experience as a transaction, which appears to be sufficient since we are not looking for sequential order of courses as compared to overall itemset frequencies. After that, the data can be used with associative classification to see course groupings that may inform how to categorize sets of classes. An initial look with an Apriori without applying thresholds reveals

C. Preprocessing Review

Given that there were so many changes to both names and course titles, it is imperative to provide those changes to people who are familiar with the courses and curriculum so they can review them. Also, given that there are two matching course descriptions in the catalog itself, that should be reviewed. And similar courses, with similar titles such as the ones that start with "Being Human" and "Becoming Human" should be reviewed. That will further help ensure that the CRISP-DM approach is followed and will allow us to spot errors and review the quality of the preprocessing steps and will further allow for adjusting to help produce a better result.

IV. Modeling Using KNIME

With preprocessing steps concluded, we now have a base model of data that can be further refined for various steps in our analysis. The algorithm and method we chose depended largely on the goals of our project. Restating those, our goals are to provide a framework to help reduce the number of credit hours from 39 to 33, define the concentration areas for elective courses, and to reduce the number of offered electives. Specifically, our data mining instructions are to focus on which electives should be retained, and to recommend sets of three or four electives that can be grouped in concentration areas.

A. The Logical Choice of Apriori

Since our data mining objectives can be achieved by determining which courses are taken the most frequently, and which courses are taken the most frequently together, the Apriori algorithm tops our list of algorithm choices for

looking for patterns in the course data. A Frequency Pattern Tree may help towards our goal, but it isn't as granular as using the Apriori algorithm with association rules and associative classification. We also aren't looking at a Generalized Sequential Pattern analysis since the goal is not about the order that courses are taken. And clustering, which is suited for numerical distance comparisons, would not be a logical choice for this project based on our textual data unless we do some further modeling for neural networks or Large Language Models (LLM).

B. Using KNIME to Examine Single Itemset Frequency

One place to start is to look at the frequency distribution of the courses. We can group by the course number to consolidate our data into lists of students and course names. From there, we can use the "Column Aggregator" node, aggregating by the lists of student names and getting an element count for the courses. This gives us an idea of the frequencies for 29 courses (after merging duplicate descriptions for 2 classes into 1 class), which is great for a preliminary look, but does not provide the context of support in terms of course combinations.

Table "default" - Rows: 29 | Spec - Columns: 3 | Properties | Flow Variables

Row ID	Course...	Course_Name	Ele...
Row5	ARTS 493	American Health Policy	143
Row28	ARTS 587	Elementary Arabic Ii	64
Row3	ARTS 491	Contemporary Pol.tho...	49
Row12	ARTS 549	Freshwater Ecology	46
Row11	ARTS 547	Communications Intern...	45
Row16	ARTS 557	19th-century British Lit...	40
Row10	ARTS 545	20th Century Russian ...	32
Row15	ARTS 555	Becoming Human	31
Row21	ARTS 569	Cell Biology And Bioche...	27
Row6	ARTS 494	Business German - A Mi...	25
Row17	ARTS 559	American South 1861-...	25
Row25	ARTS 581	Comparative Politics	23
Row22	ARTS 571 & ...	France And The Europ...	16
Row19	ARTS 565	Environmental Studies ...	14
Row2	ARTS 488	Devil's Pact Lit/film	12
Row13	ARTS 551	Aesthetics	12
Row26	ARTS 583	British Poetry 1660-1914	12
Row0	ARTS 484	Europe In A Wider World	11
Row1	ARTS 485	Evidenced Based Crime...	10
Row9	ARTS 543	Art And Religion	10
Row24	ARTS 575	Early Mesopotam Histo...	10
Row18	ARTS 561	Augustan Cultural Rev...	8
Row7	ARTS 495	Comm And The Preside...	7
Row14	ARTS 553	French Thought Since ...	7
Row8	ARTS 497	Contemp Art - 1945 To...	6
Row20	ARTS 567	Elementary German 1	6
Row23	ARTS 573	Analyzing The Pol World	6
Row4	ARTS 492	African-american Lit	1
Row27	ARTS 585	Contemporary Socio T...	1

Fig 6. Full set of single item course frequencies in KNIME after cleaning and aggregating.

C. Apriori Modeling in KNIME for Elective Categorization

Steps for further modeling the data in KNIME for Apriori include consolidating the courses into lists. If we look at each student experience as a "transaction," we can create lists of itemsets based on a student's cumulative course selections in our data. Steps for elective Apriori analysis include:

- Isolating just the 699 elective records.
- Making a threshold determination to further cut down the number of records based on attendance.
- Applying a column filter that just has the "Student Name" and "Course_Name" fields.
- Grouping by the "Student Name" variable and creating a sorted list of courses for each student as a single transaction.
- Adding a transaction identifier for each record consisting of an integer value that is unique for that transaction. This can be done using the "Math Formula" node and using the "Rownumber" function, adding one to the row number for each grouped list since the starting row number ordinal is zero.

Table "default" - Rows: 270 | Spec - Columns: 3 | Properties | Flow Variables

Row ID	Studen...	Sorted list(Course_Name)	Transa...
Row0	Al Gore	[19th-century British Literature,20th Century Ru...	1
Row1	Al Hirt	[Cell Biology And Biochemistry]	2
Row2	Al Roker	[American Health Policy]	3

Fig 7. Initial modeled transaction data ready for Apriori analysis after applying thresholds.

Prior to setting additional rules to help cut down on electives based on attendance, we can get a first glimpse of the elective frequencies and see that this data now has 270 rows of transactions in Fig. 7 and is set to be analyzed by the "Association Rule Learner (Borgelt)" node and the "Item Set Finder (Borgelt)" node in KNIME, where we can determine the support levels and how many groups of itemsets we want to create when we get to the categorization phases. Initial association rules without thresholds produce 458 rows of results.

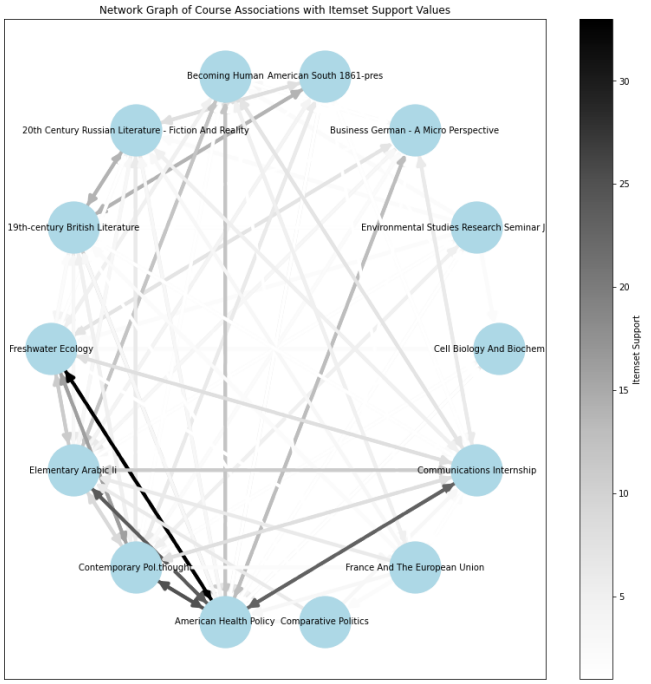


Fig 8. Network graph of Apriori results with consequents as edge nodes and support levels to other courses shown with arrows, the darker ones showing stronger itemset relationships.

In Fig. 8, a Network Graph was created using Python's Networkx package [3], with consequents around the edge of the graph and all of the listed support forming either light or dark lines where the darker lines were based on higher and repeated itemset frequencies. Strong relationships begin to emerge with the initial Apriori as "American Health Policy," "Contemporary Pol. Thought," "Elementary Arabic II," "Freshwater Ecology," and "Communications Internship" all show very dark arrows from the graphic indicating higher levels of support when those items are consequents. But it is important to note that this graphic is based on data without setting any rules for confidence and lift. Because of that, it is still too early at this stage to consider everything as relevant data as we will want to examine the single item frequencies as a starting point to cut back on the courses, and so the row count will be considerably less after cutting. For that step, we move into the evaluation phase, and afterwards will rerun the Apriori with the refined dataset.

V. EVALUATION

With preprocessing steps concluded, our first important observation can be gleaned from looking at the single course frequencies. The results reflect why we are working on this data since we can plainly see there are a number of courses with extremely low enrollment. In Table 1, applying a statistics view node to the elements reveals:

- Some courses have had only 1 student enrolled in the four years of data.
- The 50% quantile shows that half of the courses have had 12 or fewer students and average under 3 students a year.

TABLE I. COURSE ENROLLMENT FREQUENCIES OVER 4 YEARS

Metric	*Value*
Minimum	1
Maximum	143
25% Quantile	7.5
50% Quantile	12
75% Quantile	33
Mean	24.103

Where educational quality needs to be considered and results need to be reviewed by a trained eye, half of the 29 elective courses have only seen 12 students or fewer over the past 4 years, which hardly seems sustainable. The results also revealed that the 75% quantile shows enrollment of 33 students or less over the course of 4 years.

A. Rules for Keeping Courses – Single Item Frequency

As we know with Apriori, we can apply a threshold to single item frequency first since it will weed out less relevant subsets. We can consider the top 50% of elective courses that have an enrollment of greater than 12 students as a threshold, which still gives us a low baseline for our electives if you consider this is 4 years' worth of data, meaning that an elective course that has 14 students is above our threshold still averages just over 3 students a year if the course is taught annually.

S Course...	S Course_Name	I Ele...	D Top 50%
ARTS 493	American Health Policy	143	1
ARTS 587	Elementary Arabic Ii	64	1
ARTS 491	Contemporary Pol.tho...	49	1
ARTS 549	Freshwater Ecology	46	1
ARTS 547	Communications Intern...	45	1
ARTS 557	19th-century British Lit...	40	1
ARTS 545	20th Century Russian ...	32	1
ARTS 555	Becoming Human	31	1
ARTS 569	Cell Biology And Bioche...	27	1
ARTS 494	Business German - A Mi...	25	1
ARTS 559	American South 1861-...	25	1
ARTS 581	Comparative Politics	23	1
ARTS 571 & ...	France And The Europ...	16	1
ARTS 565	Environmental Studies ...	14	1

Fig 9. There are 14 remaining courses after setting a threshold of greater than 12 students enrolled over 4 years, at least having them average over 3 students a year.

We now have a modest list of 14 courses, and we'll recommend at this stage that the program keeps: "American Health Policy," "Elementary Arabic II," "Contemporary Pol. Thought," "Freshwater Ecology," "Communications Internship," "19th-century British Literature", "20th Century Russian Literature – Fiction And Reality," "Becoming Human," "Cell Biology and Biochemistry," "Business German – A Micro Perspective," "American South 1861-pres," "Comparative Politics," "France And the European Union," and "Environmental Studies" as these courses have enrollment of 14 or more.

B. Rules for Course Grouping Evaluation

Now that we've cut back on the course data and have identified courses that have had more than 12 students over the 4 years of data, we see that we have 580 relevant modeling records to use for our Apriori after joining back to the raw enrollment data using an indicator to filter out the other records. The indicator was created using a KNIME "Math Formula" node, setting an indicator of 1 for records that were above the 12-element support count level in the single item frequency list prior to joining that over to the data going thought the Apriori.

Table "default" - Rows: 580 Spec - Columns: 9 Properties Flow Variables

Row ID	S Studen...	S semest...	S Course...	S Course_Name	S Course...	S Course_Nam...	I Elemen...
Row27_Row2...	Chris Evert	Spring 2001	ARTS 549	Freshwater Ecology	ARTS 549	Freshwater Ecology	46
Row28_Row7...	Chris Evert	Spring 2002	ARTS 491	Contemporary Pol...	ARTS 491	Contemporary Pol...	49
Row29_Row9...	Chris Evert	Spring 2003	ARTS 493	American Health P...	ARTS 493	American Health P...	143
Row30_Row2...	Ruth Bader ...	Spring 2002	ARTS 549	Freshwater Ecology	ARTS 549	Freshwater Ecology	46
Row31_Row7...	Ruth Bader ...	Spring 2001	ARTS 491	Contemporary Pol...	ARTS 491	Contemporary Pol...	49
Row32_Row9...	Ruth Bader ...	Spring 2001	ARTS 493	American Health P...	ARTS 493	American Health P...	143
Row33_Row2...	Ruth Bader ...	Fall 2001	ARTS 559	American South 1...	ARTS 559	American South 1...	25
Row44_Row4...	Geraldine Fe...	Spring 2004	ARTS 587	Elementary Arabic Ii	ARTS 587	Elementary Arabic Ii	64
Row48_Row9...	CCH Pounder	Summer 2002	ARTS 493	American Health P...	ARTS 493	American Health P...	143
Row52_Row7...	CCH Pounder	Summer 2003	ARTS 491	Contemporary Pol...	ARTS 491	Contemporary Pol...	49
Row54_Row2...	CCH Pounder	Summer 2001	ARTS 545	20th Century Rus...	ARTS 545	20th Century Rus...	32

Fig 10. There are 580 remaining rows after filtering out the lower 50% quantile of the records based on lower attendance.

For the updated Apriori, we are using the "Association Rule Learner (Borgelt)" node with a default minimum support of 3.0, as described in Section VI., and we now have 458 rows of course groupings, down from 580, after removing less frequent itemsets with support of 12 or less. A histogram (see Fig. 11) in KNIME with 6 bins of support groupings gives us an idea of what the Apriori our data looks like when considering each student experience as a transaction and weeding out the less frequent itemsets. A quick look shows us

that the automatically chosen by KNIME support bin range of 16.505 to 20.34 has 159 transactions with support inside that range, which is our largest grouping when binning the data into 6 bins.

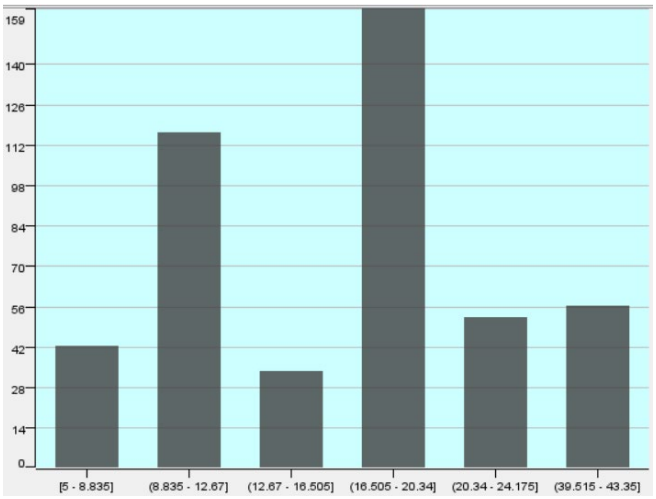


Fig 11. KNIME histogram of 6 bins of support groupings with frequency counts of Apriori itemsets on the y-axis after cutting items with a support of 12 or less.

Now we can examine association rules to determine thresholds using essential attributes, such as confidence and lift, to help us in our task of identifying some of the more likely course groupings and associations that allow us categorize them.

1) Course Grouping Thresholds: Keeping with the support level of 3, we can weed out a number of itemsets produced by the Apriori, taking us down to 164 rows of data. From there, we can determine our best groupings by examinging we our objective, which dictates that we are looking for groupings of three or four courses that will help us make recommendations for elective categories, so we will cut out groupings under 3 elements in each set. This requires getting a count of the consequent plus the antecedent groupings. To help with this task, information was available on the KNIME Community Forum [4] on how to accomplish this, and the Vernalis extension that has a "Collection Size" node that counts items in a collection, was installed and used to count the antecedents in the Apriori output. Adding one to the counts allows also considers the antecedent. This leaves us with 91 rows of groupings that are above the three or more threshold.

Table "default" - Rows: 91 Spec - Columns: 11 Properties Flow Variables

Row ID	S Consequent	[...] Antecedent	D Coiurse Group Count
Row50	Elementary Arabic Ii	[France And The European Union,Becoming Hum...	3
Row51	Becoming Human	[France And The European Union,Elementary Ar...	3
Row52	France And The E...	[Becoming Human,Elementary Arabic Ii]	3
Row141	American Health P...	[Business German - A Micro Perspective,Commun...	3
Row142	Communications I...	[Business German - A Micro Perspective,America...	3
Row143	Business German -...	[Communications Internship,American Health Poli...	3
Row158	American Health P...	[Business German - A Micro Perspective,Freshwa...	3

Fig 12. There are 91 remaining Apriori rows that have a minimum of 3 courses per group.

2) Additional Thresholds For Qualified Results: At this stage, we have many attributes for courses and course combinations that were created by the "Association Rule Learner (Borgelt)" node in KNIME. Those key attributes include:

- **ItemSetSupport:** This is the raw count of how often course combinations occur together with the consequent course and antecedent course and courses. This helps us see less common course combinations.
- **RelativeItemSetSupport%:** This converts the itemset support into a percentage of the total number of course combinations. It will allow us to compare the popularity of course groupings to the other course groupings in the dataset.
- **RuleConfidence%:** This measures the likelihood that if a student enrolls in one course, they will also take another course or course grouping. It will allow us to determine the strength of the relationship between courses.
- **RuleLift:** This allows us to look at how likely courses are taken together if the enrollment was independent. It is useful finding unique course associations that may not be obvious at first glance.
- **RuleLift%:** The is the percentage likelihood that courses are taken together, where amounts over 100 % mean they are extremely likely. This is a version of RuleLift that allows us to say that someone is a percentage likely to take a course given that they've taken a course or course grouping.

We wouldn't want to consider any groupings that do not have a lift of 1 or more, as 1 would be the baseline for how likely it is that courses are taken together. Applying that threshold using a "Row Filter" node, we now have just 80 groupings remaining.

Table "default" - Rows: 80 Spec - Columns: 8 Properties Flow Variables

Row ID	S Consequent	[...] Antecedent	D Rul...
Row211	American South 1...	[19th-century British Literature,Contemporary P...	1,080
Row216	American South 1...	[19th-century British Literature,Contemporary P...	648
Row179	19th-century Briti...	[American South 1861-pres,20th Century Russia...	605.77
Row338	Communications I...	[20th Century Russian Literature - Fiction And R...	600
Row209	Contemporary Pol...	[American South 1861-pres,19th-century British ...	551.02
Row181	American South 1...	[20th Century Russian Literature - Fiction And R...	540
Row222	American South 1...	[19th-century British Literature,Elementary Arabi...	540
Row210	19th-century Briti...	[American South 1861-pres,Contemporary Pol.th...	519.23
Row307	20th Century Rus...	[19th-century British Literature,Contemporary P...	506.25
Row240	American South 1...	[Contemporary Pol.thought,Elementary Arabic Ii]	480

Fig 13. There are 80 remaining Apriori rows that have a lift of 1 or higher after removing courses that have 12 or fewer students over the 4-year data collection period with groupings of 3 or more and support of at least 3.

Additionally, we wouldn't want results that have too low of a confidence score. Setting a 50% threshold for "RuleConfidence%" means we'll remove those weaker results that are correct less than half the time. This will allow us to concentrate on the more significant and reliable patterns in the data. Applying that filter now leaves us with 43 rows of data.

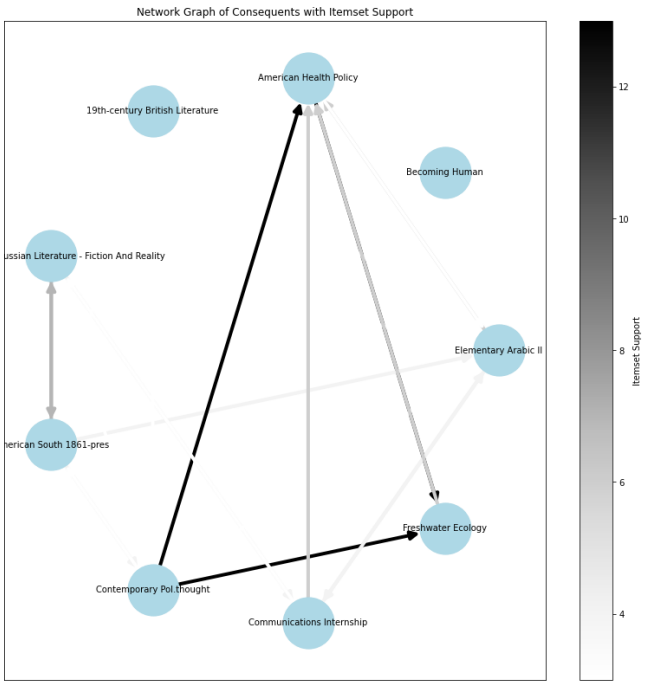


Fig 14. Network graph showing the remaining consequents and support to antecedents.

The network graph in Fig. 14 reveals the strong relationships based on itemset counts between the consequents and antecedents for the 43 association rules that remain. The graph also points to further cuts if they are needed in follow up, but we can pause because we have winnowed down our data significantly. With this small set of data remaining, we now can find distinct groupings since we do not have 43 unique sets of courses as the Apriori swaps out the antecedent and consequents and some of the same groupings appear more than once. In other words, there are a significant number of duplicate groups created by the Apriori algorithm that score them more than once with our results. To help identify these groupings, we can do the following steps in KNIME:

- Concatenate the antecedent and consequent using the "String Manipulation" node that is a full group.
- Split the full group out using the "Cell Splitter" to create separate columns.
- Use a "Java Snippet" node to help sort the columns and append them back together. As the Apriori tests different consequents against the antecedents, it switches the order of these and scores them again. We can create an array for the three courses in the groups and sort them simply since Java has an "Arrays.sort()" function [5]. Then they can be reassembled and replace the full group variable that already exists. This will create an alpha ordered list that we can compare against the other alpha sorted lists for duplication.
- Use a "Group By" node to give us a deduped list.

```
// Enter your code here:
String[] courses = new String[4];
courses[0] = c_FullGroup_Arr0;
courses[1] = c_FullGroup_Arr1;
courses[2] = c_FullGroup_Arr2;
if (c_FullGroup_Arr3.length()>0)
{
  courses[3] = ", " + c_FullGroup_Arr3;
}
else
{
  courses[3] = "";
}
java.util.Arrays.sort(courses);
out_FullGroup =  courses[0] + ", " + courses[1] + ", " + courses[2] + courses[3];
out_FullGroup = out_FullGroup.substring(2);
```

Fig 15. The "Java Snippet" in KNIME can help with ordering array data by breaking it apart and reassembling it.

Using the "Group By" node also allows us to choose the maximum "RuleLift," maximum "RuleConfidence%," and cumulative "Itemsupport" or each set, as well as count the number of instances of the groupings. We can now see we have 28 unique groupings remaining in Fig. 16.

Table "default" - Rows: 28 Spec - Columns: 5 Properties Flow Variables

Row ID	Full Group	Max*(R...	Max*(R...	Sum(It...	Count(...
Row0	19th-century British Literature, 20th Century Russian Literature - Fiction And RealityAmerican South 1861-pres	6.058	87.5	21	3
Row1	19th-century British Literature, 20th Century Russian Literature - Fiction And RealityContemporary Pol.thought	5.062	60	6	2
Row2	19th-century British Literature, American Health PolicyElementary Arabic II	1.154	50	3	1
Row3	19th-century British Literature, American South 1861-presContemporary Pol.thought	6.48	60	6	2
Row4	19th-century British Literature, American South 1861-presElementary Arabic II	5.4	50	6	2
Row5	19th-century British Literature, Contemporary Pol.thoughtElementary Arabic II	2.755	60	6	2
Row6	20th Century Russian Literature - Fiction And Reality, American Health PolicyCommunications Internship	1.385	60	3	1
Row7	20th Century Russian Literature - Fiction And Reality, American Health PolicyContemporary Pol.thought	1.154	50	3	1
Row8	20th Century Russian Literature - Fiction And Reality, American Health PolicyElementary Arabic II	1.385	60	3	1
Row9	20th Century Russian Literature - Fiction And Reality, Communications InternshipElementary Arabic II	4.8	80	8	2
Row10	20th Century Russian Literature - Fiction And Reality, Contemporary Pol.thoughtElementary Arabic II	3.306	60	6	2
Row11	American Health Policy, 20th Century Russian Literature - Fiction And Reality, Communications InternshipEle...	6	100	9	3
Row12	American Health Policy, Becoming HumanCommunications Internship	1.319	57.1	4	1
Row13	American Health Policy, Becoming HumanFreshwater Ecology	1.385	60	3	1
Row14	American Health Policy, Business German - A Micro PerspectiveCommunications Internship	1.154	50	3	1
Row15	American Health Policy, Business German - A Micro PerspectiveElementary Arabic II	1.154	50	3	1
Row16	American Health Policy, Business German - A Micro PerspectiveFreshwater Ecology	1.978	85.7	6	1

Fig 16. We are down to 28 groupings of elective courses for our course grouping categorization after aggregating by unique course sets.

C. *Remaining Course Groups – Our Course Categories*

After careful application of Apriori algorithm and setting thresholds for lift and confidence, we have identified 28 unique groupings of elective courses. These groupings represent strong patterns in student elective choices, suggesting potential concentration areas within the Arts program.

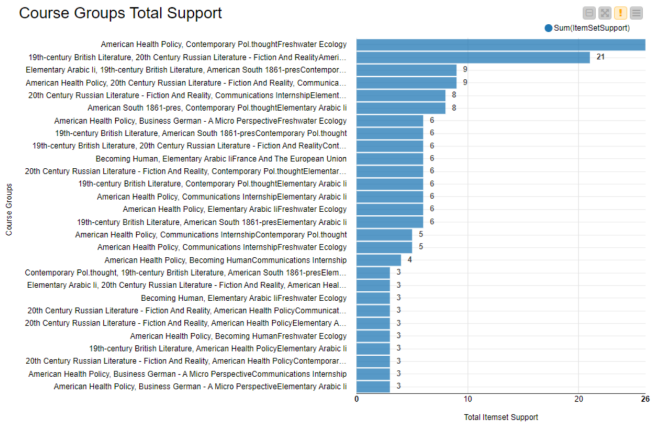


Fig 17. We are left with 28 course groupings seen in descending order by cumulative itemset support.

Each grouping combines courses that share thematic or disciplinary connections, reflecting student interests and potential interdisciplinary pathways. For example:

- **Grouping "19th-century British Literature," "20th Century Russian Literature – Fiction And Reality," and "Communications Internship":** This grouping, with a rule lift of 6.058 and confidence of 87.5%, represents a strong intersection

of cultural studies and practical communication skills. The high metrics suggest that students interested in a deep dive into different literary traditions also value practical applications of their communication skills, possibly in careers related to media, education, or cultural diplomacy. This could inform the development of an interdisciplinary course set that integrates literary studies with media and communication, preparing students for diverse roles that require both analytical and practical skills.

- **Grouping "American Health Policy," "20th Century Russian Literature – Fiction And Reality," and "Freshwater Ecology":** With a rule lift of 6 and a confidence of 100%, this diverse set of courses indicates a unique blending of health policy, environmental science, and cultural studies. The perfect confidence score implies that students taking one of these courses are highly likely to take the others, suggesting a study program that might appeal to students interested in global health issues, environmental policies, and the socio-political contexts of different regions. This grouping could lead to the creation of a new interdisciplinary track focusing on global environmental health policies.

- **Grouping "Elementary Arabic II," "19th-century British Literature," and "American Health Policy":** This grouping showcases an extremely high rule lift of 10.8 and confidence of 100%, suggesting a significant correlation among students studying these seemingly disparate subjects. The combination of language studies, literature, and health policy could reflect a student cohort preparing for roles that combine cultural competence, historical knowledge, and policy-making, such as international relations or global health. This insight could guide the development of a specialized program that equips students with the skills to operate effectively at the intersection of culture, history, and policy.

VI. Summary and Recommendations

Overall, this analysis has revealed critical insights into the enrollment patterns of Arts program electives, highlighting the necessity for a comprehensive review and strategic realignment of the curriculum. By employing the Apriori algorithm and association rules under the structured CRISP-DM framework, we have identified both popular and under-enrolled courses, which informs our recommendations for curriculum optimization. Notably, the data points to a significant concern: the majority of electives show very low enrollment numbers, with half of the courses attracting 12 or fewer students over 4 years, which raises questions about financial and educational viability. This insight truly highlights the need for further financial and sustainability analysis.

Furthermore, it is essential to implement standardized course naming conventions across the university. While data mining facilitates the standardization of data for analytical purposes, it does not address the broader need for consistency in how courses are marketed and communicated. To ensure both clarity and efficiency in data management and external communications, the university administration and faculty should collaborate to refine and standardize course names. This process should include a thorough review of existing course titles to correct typographical errors and standardize punctuation and abbreviations, which will enhance data integrity and consistency across university records. Moreover, establishing these standards is not only critical for effective data analysis, but is also crucial for maintaining the institution's professional image and ensuring accurate communication within and outside the university. We recommend that the university administration undertakes a systematic review and standardization of course names to support operational effectiveness.

A. Data Clearning and Client Review Considerations

The initial phase of our study focused on preparing a clean, analyzable dataset. This process highlighted the importance of addressing missing values, duplicates, and inconsistent course naming conventions. Critical to this was the realization that further review with clients is essential, especially for decisions around course name standardizations and understanding the implications of duplications. These considerations are foundational to ensuring that subsequent studies accurately reflect the curriculum's current state.

B. Course Frequency Analysis and Sustainability

Our analysis revealed a significant portion of courses with low enrollment, suggesting that half of the courses with 12 or fewer students enrolled over 4 years have an average of 3 students per course per year may not be sustainable. This finding alone indicates a need for a deeper exploration into the program's financial objectives and broader impacts on the university. A formal cost-benefit analysis should be conducted to determine the financial sustainability of each course, considering not only enrollment but also operational costs.

C. Strengths of the Recommended Courses

The courses that met our criteria for retention are characterized by their interdisciplinary nature and global perspective, combining fields such as health, communication, ecology, politics, classic literature, and Arabic language. This diverse curriculum aligns with contemporary global challenges and positions the Arts program as a leader in delivering a well-rounded, relevant educational experience.

D. CRISP-DM Next Steps

Moving forward, we can continue to use CRISP-DM and follow up with some key steps*:*

- **Business Understanding:** Engage with stakeholders to reassess and possibly realign program objectives, considering both quantitative findings and qualitative aspects of course value.
- **Data Understanding and Preparation:** Collaborate with clients to validate preprocessing decisions and explore data beyond simple frequency and attendance, potentially incorporating course evaluations, academic outcomes, and labor market trends.
- **Modeling and Evaluation:** Continue to refine our models based on stakeholder feedback, exploring alternative metrics that might better capture course value and student engagement.
- **Deployment:** Implement recommendations in a phased approach, closely monitoring the impact on

enrollment and student satisfaction to adjust course offerings if needed.

E. Study Limitations

The dataset, covering 4 years of curriculum data, contained significant typos and inconsistencies, requiring extensive preprocessing that could introduce biases. The absence of additional data on ideal class sizes and financial sustainability limits the scope of our recommendations to enrollment trends without considering economic feasibility. These limitations should be considered in the interpretation and application of our results, and holdout samples could be used for sensitivity analysis in follow-up work.

F. Conclusion

This investigation has given Arts program administrators advanced insights essential for the curriculum's modernization. Our analysis not only highlights the interdisciplinary and global strengths of the program, but identifies critical areas for improvement in course offerings and administrative practices.

As we move forward, we strongly advocate for the continuous standardization of course naming and descriptions. This standardization is crucial not only to facilitate communication and maintain data integrity, but also to ensure that our analyses are based on reliable data. Moreover, integrating a detailed cost-benefit analysis is vital to assess the financial viability of each course offering, aligning our academic objectives for economic sustainability.

Further investigation should be undertaken to delve deeper into the data and uncover additional insights to address evolving educational trends and financial constraints. By using strategies informed by the CRISP-DM, we can ensure that the Arts program remains at the cutting edge of educational development, offering valuable and sustainable educational experiences that meet both current and future student needs.

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